2. Quadratic Equations

Q11. Solve by factorisation: $x^2 - 7x + 12 = 0$.
Given: $x^2 - 7x + 12 = 0$
Factorise: $x^2 - 7x + 12 = (x - 3)(x - 4)$
$(x - 3)(x - 4) = 0 \Rightarrow x = 3$ or $x = 4$
Answer: $x = 3, 4$.

Q12. Solve using quadratic formula: $3x^2 + 4x - 5 = 0$.
Given: $3x^2 + 4x - 5 = 0 \Rightarrow a = 3, b = 4, c = -5$
Quadratic formula: $x = [-b \pm \sqrt{(b^2 - 4ac)}] / (2a)$
$D = b^2 - 4ac = 16 - 4(3)(-5) = 16 + 60 = 76$
$x = [-4 \pm \sqrt{76}] / 6 = [-4 \pm 2\sqrt{19}] / 6$
$\therefore x = (-2 \pm \sqrt{19})/3$
Answer: $x = (-2 + \sqrt{19})/3$ or $x = (-2 - \sqrt{19})/3$.

Q13. Solve: $2x^2 - 5x + 3 = 0$ and hence find the sum and product of its roots.
Given: $2x^2 - 5x + 3 = 0$
Factorise: $2x^2 - 5x + 3 = (2x - 3)(x - 1)$
$(2x - 3)(x - 1) = 0 \Rightarrow x = 3/2$ or $x = 1$
Sum of roots $= 3/2 + 1 = 5/2$
Product of roots $= (3/2)\times 1 = 3/2$
Answer: Roots = 3/2 and 1; Sum = 5/2; Product = 3/2.

Q14. Find the discriminant of $5x^2 - 6x + 2 = 0$ and hence comment on the nature of its roots.
Given: $5x^2 - 6x + 2 = 0 \Rightarrow a = 5, b = -6, c = 2$
$D = b^2 - 4ac = (-6)^2 - 4(5)(2) = 36 - 40 = -4$
Since $D < 0$, roots are not real (complex conjugate).
Answer: Discriminant = −4; roots are not real.

Q15. The product of two consecutive positive integers is 506. Form a quadratic equation and find the integers.
Let integers be n and n + 1.
Given: $n(n + 1) = 506 \Rightarrow n^2 + n - 506 = 0$
$D = 1 + 4\times 506 = 2025 = 45^2$
$n = [-1 \pm 45]/2 \Rightarrow n = 22$ or $n = -23$
Consecutive positive integers ⇒ 22 and 23.
Answer: 22 and 23.

Q16. The sides of a rectangular park are in the ratio 3:2 and its area is 1500 m². Find the length and breadth of the park using a quadratic equation.
Let length = 3x and breadth = 2x.
Area = 1500 ⇒ $(3x)(2x) = 1500 \Rightarrow 6x^2 = 1500$
$x^2 = 250 \Rightarrow x = \sqrt{250} = 5\sqrt{10}$ (positive)
Length $= 3x = 15\sqrt{10}$ m, Breadth $= 2x = 10\sqrt{10}$ m
Answer: Length $= 15\sqrt{10}$ m, Breadth $= 10\sqrt{10}$ m.

Q17. A pole is broken such that its top touches the ground and makes a triangle with the ground. The point of break is 5 m above the ground and the top of the pole touches the ground at a distance of 12 m from the foot of the pole. Find the original height of the pole.
Lower unbroken part = 5 m (vertical). Distance from foot to touching point = 12 m (horizontal).
Upper broken part is the hypotenuse of a right triangle.
Upper part $= \sqrt{(5^2 + 12^2)} = \sqrt{(25 + 144)} = \sqrt{169} = 13$ m
Original height = 5 + 13 = 18 m
Answer: Original height of pole = 18 m.

Q18. If α and β are the roots of $x^2 - 4x + 3 = 0$, form the quadratic equation whose roots are 2α and 2β.
For $x^2 - 4x + 3 = 0$: $\alpha + \beta = 4$ and $\alpha\beta = 3$
New roots: 2α and 2β
Sum $= 2(\alpha + \beta) = 8$
Product $= (2\alpha)(2\beta) = 4\alpha\beta = 12$
Required equation: $x^2 - (\text{sum})x + (\text{product}) = 0$
$\therefore x^2 - 8x + 12 = 0$
Answer: $x^2 - 8x + 12 = 0$.

Q19. Solve for x: $1/x + 1/(x - 2) = 3/4$.
Given: $1/x + 1/(x - 2) = 3/4$; $x \neq 0, 2$
LCM $= x(x - 2)$
$(x - 2 + x) / [x(x - 2)] = 3/4 \Rightarrow (2x - 2)/[x(x - 2)] = 3/4$
Cross-multiply: $4(2x - 2) = 3x(x - 2)$
$8x - 8 = 3x^2 - 6x \Rightarrow 3x^2 - 14x + 8 = 0$
Factorise: $3x^2 - 14x + 8 = (3x - 2)(x - 4)$
$\therefore x = 2/3$ or $x = 4$ (both valid since $x \neq 0,2$)
Answer: $x = 4$ or $x = 2/3$.

Q20. The sum of the reciprocals of two consecutive natural numbers is 11/30. Find the numbers by forming and solving a quadratic equation.
Let numbers be n and n + 1.
$1/n + 1/(n + 1) = 11/30$
$(2n + 1)/[n(n + 1)] = 11/30$
$30(2n + 1) = 11n(n + 1)$

$60n + 30 = 11n^2 + 11n \Rightarrow 11n^2 - 49n - 30 = 0$

$D = (-49)^2 - 4(11)(-30) = 2401 + 1320 = 3721 = 61^2$

$n = [49 \pm 61]/22 \Rightarrow n = 5$ or $n = -6/11$ (reject)

Therefore numbers are 5 and 6.

Answer: 5 and 6.